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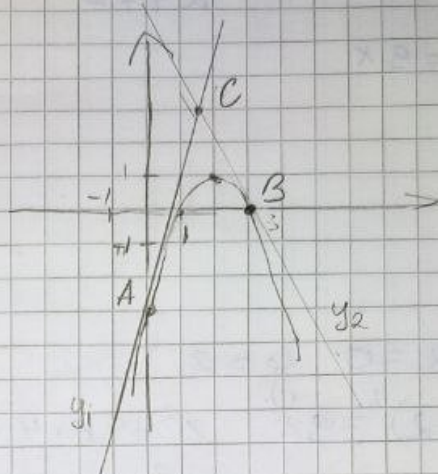
(1)

$$y = -x^2 + 4x - 3$$

$$A(0, -3) \quad B(3, 0)$$

$$A \in y_1' \quad \cup \quad B \in y_2'$$

S = ?



$$y = -(x-2)^2 + 1$$

$$y' = -2(x-2) = -2x + 4$$

$$x_a = 0$$

$$K_1 = -2 \cdot 0 + 4 = 4$$

$$y_1 = K_1 \cdot x + b$$

$$y_1 = 4 \cdot 0 + b \quad y_a = 4 \cdot 0 + b \Rightarrow -3 = b$$

$$y_1 = 4x - 3$$

$$y_2 = K_2 x + b$$

$$x_b = 3$$

$$K_2 = -2 \cdot 3 + 4 = -2$$

$$y_2 = -2 \cdot x + b$$

$$y_b = -2x_b + b = 0$$

$$-2 \cdot 3 + b = 0$$

$$b = 6$$

$$]C \in y_1, y_2 :$$

$$y_1 = y_2$$

$$4x_c - 3 = -2x_c + 6$$

$$6x_c = 9$$

$$x_c = \frac{3}{2}$$

$$y_c = -2 \cdot \frac{3}{2} + 6 = 3 \quad C\left(\frac{3}{2}, 3\right)$$

$$\begin{aligned} S &= \int_0^{\frac{3}{2}} (y_1 - y) dx + \int_{\frac{3}{2}}^3 (y_2 - y) dx = \int_0^{\frac{3}{2}} (4x - 3 + x^2 - 4x + 3) dx + \int_{\frac{3}{2}}^3 (-2x + 6 + x^2 - 4x + 3) dx \\ &= \int_0^{\frac{3}{2}} x^2 dx + \int_{\frac{3}{2}}^3 (x^2 - 6x + 9) dx = \left. \frac{x^3}{3} \right|_0^{\frac{3}{2}} + \left. \left(\frac{x^3}{3} - 6 \frac{x^2}{2} + 9x \right) \right|_{\frac{3}{2}}^3 \\ &= \left(\frac{3}{2} \right)^3 \cdot \frac{1}{3} - \frac{0}{3} + \frac{3^3}{3} - 6 \frac{3^2}{2} + 9 \cdot 3 - \left(\left(\frac{3}{2} \right)^3 \cdot \frac{1}{3} - 6 \cdot \left(\frac{3}{2} \right)^2 \cdot \frac{1}{2} + 9 \cdot \frac{3}{2} \right) = \\ &= \cancel{9/8} + 9 - \cancel{27} + \cancel{27} - \cancel{9/8} + \frac{27}{4} - \frac{27}{2} = 9 - \frac{27}{4} = \frac{36 - 27}{4} = \frac{9}{4} \end{aligned}$$

②

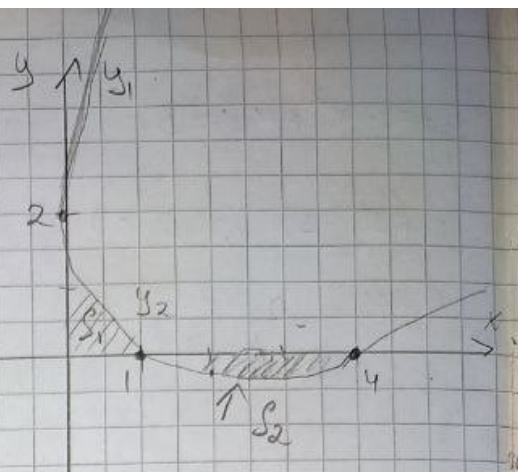
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$$(y - x - 2)^2 = 9x$$

$$y_1 = 0$$

$$y_2 = 0$$

$$x \geq 0$$



$$x=0: y-2=0 \quad y=2$$

$$y=0: (x+2)^2 = 9x \quad x^2 + 4x + 4 - 9x = 0$$

$$x^2 - 5x + 4 = 0$$

$$(x-1)(x-4) = 0$$

$$x=1 \quad x=4$$

$$(y - x - 2)^2 = 9x \Leftrightarrow \begin{cases} y_1 = x + 2 + 3\sqrt{x} \\ y_2 = x + 2 - 3\sqrt{x} \end{cases}$$

x	1	2	9
y ₁	6	8,24	20
y ₂	0	-0,24	2

$$S_1 = \int_0^1 y_2 dx = \int_0^1 (x - 3\sqrt{x} + 2) dx = \left[\frac{x^2}{2} - 3 \cdot \frac{2}{3} \cdot x^{\frac{3}{2}} + 2x \right]_0^1 =$$

$$= \frac{1}{2} - 2 + 2 = \frac{1}{2}$$

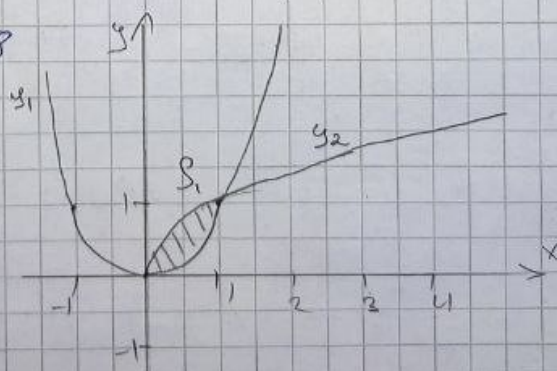
$$S_2 = \int_1^4 (0 - y_2) dx = - \left[\frac{x^2}{2} + 2x^{\frac{3}{2}} - 2x \right]_1^4 = - \left[\frac{4^2}{2} + 2 \cdot 8 - 2 \cdot 4 - \left(-\frac{1}{2} + 2 - 2 \right) \right] = -8 + 16 - 8 + \frac{1}{2} = \frac{1}{2}$$

$$S_1 + S_2 = \frac{1}{2} + \frac{1}{2} = 1$$

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$$y_1 = x^2$$

$$y_2 = \sqrt{x}$$



$$S_1 = \int_0^1 (y_2 - y_1) dx = \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} - \frac{x^3}{3} \right]_0^1 = \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{1}{3} x^3 \right]_0^1 =$$

$$= \frac{2}{3} \cdot 1 - \frac{1}{3} \cdot 1 - \frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 0 = \frac{1}{3}$$

(3)

$$x = a(t - \sin t)$$

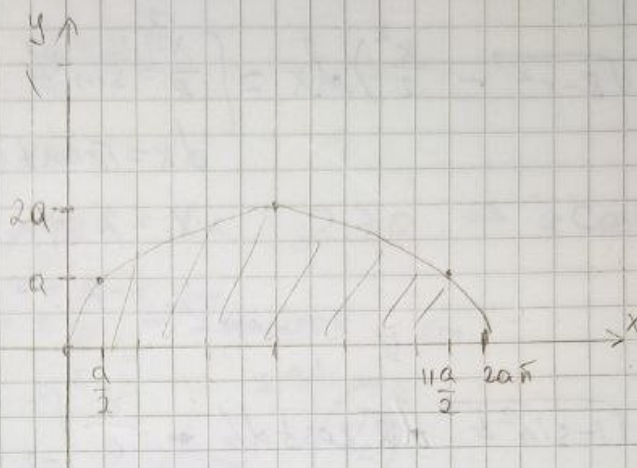
$$y = a(1 - \cos t)$$

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$$y=0$$

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
x	0	$a/2$	πa	$11a/2$	$2a\pi$
y	0	a	2a	a	0

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$$\begin{aligned} S &= \int_0^{2\pi} y_t dx_t = \int_0^{2\pi} y_t x'_t dt = \int_0^{2\pi} a(1 - \cos t) \cdot a(1 - \cos t) dt = \\ &= \int_0^{2\pi} a^2 (1 - 2\cos t + \cos^2 t) dt = a^2 \left(t - 2\sin t + \int_0^{2\pi} \frac{1 + \cos 2t}{2} dt \right) \Big|_0^{2\pi} = \\ &= a^2 t - 2a^2 \sin t + \frac{a^2}{2} \left(t + \sin 2t \cdot \frac{1}{2} \right) \Big|_0^{2\pi} = a^2 \left(t - 2\sin t + \frac{t}{2} + \frac{\sin 2t}{2} \right) \Big|_0^{2\pi} = \\ &= a^2 \left(\frac{3}{2} \cdot 2\pi - 2\sin 2\pi + \frac{1}{2} \sin 4\pi - \frac{1}{2} \cdot 0 - 2\sin 0 + \frac{1}{2} \sin 0 \right) = \\ &= a^2 (\pi + 2\pi) = 3\pi a^2 \end{aligned}$$

$$x^2 + y^2 = 8$$

$$y = \frac{x^2}{2}$$

$$S_1 \cup S_2 = ?$$

$$S_{\text{oduz}} = \pi \cdot 8^2 = 8\pi$$

$$x^2 + \left(\frac{x^2}{2}\right)^2 = 8$$

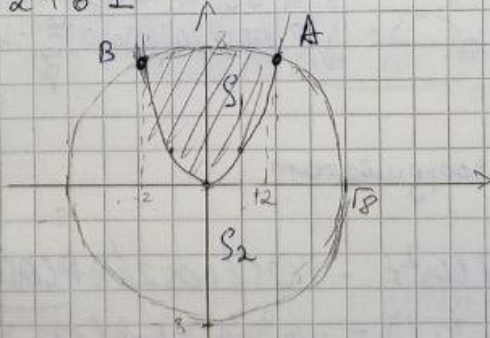
$$t + \frac{1}{4}t^2 = 8$$

$$\begin{cases} t = x^2, t \geq 0 \\ t^2 + 4t - 32 = 0 \end{cases}$$

$$(t-4)(t+8) = 0$$

$$\begin{cases} t = -8 - \sqrt{16}, t \geq 0 \\ t = 4 \Rightarrow x = \pm 2 \end{cases}$$

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$$\textcircled{4} \quad y_a = \frac{2^2}{2} = 2 \quad x^2 + y^2 = 8 \Rightarrow \begin{cases} y = \sqrt{8-x^2} - \text{верх} \\ y = -\sqrt{8-x^2} - \text{низ} \end{cases}$$

$$y_b = \frac{(-2)^2}{2} = 2$$

$$S_1 = \int_{-2}^2 (\sqrt{8-x^2} - \frac{x^2}{2}) dx = \left[\begin{matrix} \frac{x^2}{8} = \sin^2 t & x = \sqrt{8} \sin t \\ dx = \sqrt{8} \cos t dt & t = \arcsin \frac{x}{\sqrt{8}} \end{matrix} \right] =$$

$$x_1 = 2 \Rightarrow t_1 = \arcsin \frac{1}{2} = \frac{\pi}{4}$$

$$x_0 = -2 \Rightarrow t_0 = \arcsin \frac{-1}{2} = -\frac{\pi}{4}$$

$$= \int_{-\pi/4}^{\pi/4} (\sqrt{8} \cdot \sqrt{1-\sin^2 t} - 8 \cos^2 t) \sqrt{8} \cos t dt + \frac{x^3}{6} \Big|_{-2}^2 = 8 \int_{-\pi/4}^{\pi/4} \cos^2 t dt - \left(\frac{8}{6} - \frac{-8}{6} \right) =$$

$$= 2 \cdot 2 \int_{-\pi/4}^{\pi/4} \frac{1+\cos 2t}{2} dt - \frac{8}{3} = \left(2 \cdot \frac{4t}{2} + 2 \sin 2t \right) \Big|_{-\pi/4}^{\pi/4} - \frac{8}{3} =$$

$$= \left(4 \cdot \frac{\pi}{4} + 2 \sin \frac{\pi}{2} + 4 \cdot \frac{\pi}{4} - 2 \sin \frac{-\pi}{2} \right) - \frac{8}{3} = 2\pi + 2 + 2 - \frac{8}{3} =$$

$$2\pi + 4 - \frac{2}{3} = 2\pi + \frac{10}{3} = \frac{4}{3} + 2\pi$$

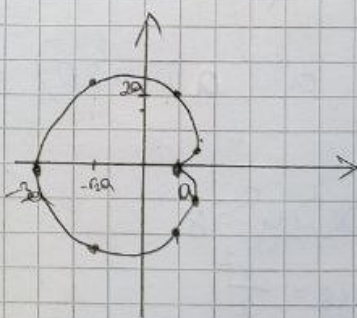
$$S_2 = S_{\text{outer}} - S_1 = 8\pi - 2\pi - \frac{4}{3} = 6\pi - \frac{4}{3}$$

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$$x = 2a \cos t - a \cos 2t$$

$$y = 2a \sin t - a \sin 2t$$

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$
x	a	$\frac{3a}{2}$	a	$\frac{a}{2}$	-a	$-\frac{3a}{2}$	-a	$\frac{a}{2}$
y	0	$\frac{a}{2}$	a	$\frac{a}{2}$	0	$-\frac{a}{2}$	-a	$-\frac{a}{2}$



Перейдем к полярным координатам

$$r^2 = x^2 + y^2 = a^2 (4 \cos^2 t - 4 \cos t \cos 2t + \cos^2 2t) + a^2 (4 \sin^2 t - 4 \sin t \sin 2t + \sin^2 2t) = 4a^2 + a^2 - 4a^2 (\cos t \cos 2t + 2 \sin^2 t \cos t) =$$

$$= 5a^2 - 4a^2 \cos t (\cos^2 t - \sin^2 t + 2 \sin^2 t) = 5a^2 - 4a^2 \cos t$$

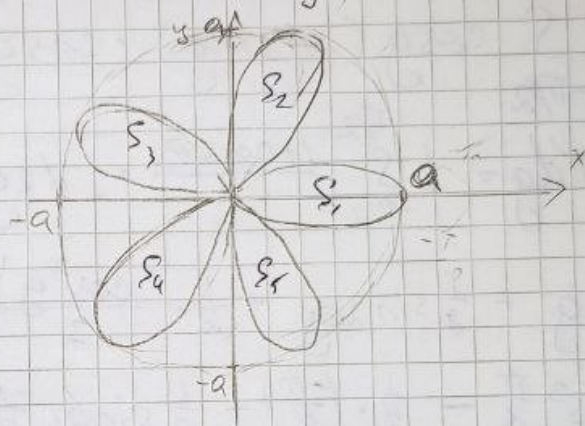
$$S = \frac{1}{2} \int_{\varphi_0}^{\varphi_1} r^2 d\varphi = \frac{1}{2} \int_0^{2\pi} (5a^2 - 4a^2 \cos t) dt = \frac{a^2}{2} \int_0^{2\pi} (5 - 4 \cos t) dt =$$

$$= \frac{a^2}{2} (5t - 4\sin t) \Big|_0^{2\pi} = \frac{a^2}{2} (5 \cdot 2\pi - 4\sin 2\pi - 5 \cdot 0 + 4\sin 2 \cdot 0) \stackrel{5}{=} \\ = 5a^2\pi$$

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$$\rho = a \cos 5\varphi$$

ρ - радиус полярной оси, т.к. $\rho \geq 0 \Rightarrow a \cos 5\varphi \geq 0$
 $-\frac{\pi}{2} \leq 5\varphi \leq \frac{\pi}{2} \Rightarrow -\frac{\pi}{10} \leq \varphi \leq \frac{\pi}{10}$ - Интервал из 5-ти



$$S_1 = S_2 = S_3 = S_4 = S_5$$

$$S = 5S_1 = 5 \cdot \frac{1}{2} \int_{-\pi/10}^{\pi/10} a^2 \cos^2 5\varphi d\varphi = \frac{5}{2} a^2 \int_{-\pi/10}^{\pi/10} \cos^2 5\varphi d\varphi = \frac{5a^2}{2} \int_{-\pi/10}^{\pi/10} \frac{\cos 10\varphi + 1}{2} d\varphi = \\ = \frac{5a^2}{4} \cdot \frac{1}{10} \int_{-\pi/10}^{\pi/10} (1 + \cos 10\varphi) d10\varphi = \left(\frac{5a^2}{4} \cdot \frac{1}{10} \cdot 10\varphi + \frac{5a^2}{4} \cdot \frac{1}{10} \cdot \sin 10\varphi \right) \Big|_{-\pi/10}^{\pi/10} = \\ = \frac{5a^2}{4} \left(\frac{\pi}{10} + \frac{1}{10} \sin \frac{\pi}{10} \cdot 10 - \left(-\frac{\pi}{10} + \frac{1}{10} \sin \frac{-\pi}{10} \cdot 10 \right) \right) = \frac{5a^2}{4} \left(\frac{\pi}{10} + \frac{\pi}{10} \right) = \\ = \frac{5a^2}{4} \cdot \frac{\pi}{5} = \frac{a^2 \pi}{4}$$

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$$\rho = \sqrt{1-t^2} \rightarrow |t| \leq 1 \rightarrow -1 \leq t \leq 1$$

$$\varphi = \arcsin t + \sqrt{1-t^2}$$

$$S = \frac{1}{2} \int_{-1}^1 \rho^2 \varphi' dt = \frac{1}{2} \int_{-1}^1 (1-t^2) \left(\frac{1}{\sqrt{1-t^2}} + \frac{1}{2\sqrt{1-t^2}} \cdot (-2t) \right) dt =$$

$$\textcircled{6} = \frac{1}{2} \int_{-1}^1 \frac{(1-t^2)}{1} \cdot \frac{1-t}{1-t^2} \cdot dt = \frac{1}{2} \int_{-1}^1 (1-t^2) \cdot (1-t) dt \quad \textcircled{=}$$

Заменим $t = \sin g$ $dt = \cos g dg$
 $\sqrt{1-t^2} = \cos g$

$$t_1 = 1 \rightarrow g_1 = 1 \Rightarrow g_1 = \frac{\pi}{2}$$

$$t_0 = -1 \rightarrow \sin g_0 = -1 \Rightarrow g_0 = -\frac{\pi}{2}$$

$$\begin{aligned} \textcircled{=} & \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos g (1 - \sin g) \cdot \cos g dg = \frac{1}{2} \int_{-\pi/2}^{\pi/2} (\cos^2 g - \cos^2 g \cdot \sin g) dg = \\ & \frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2g}{2} \cdot \frac{1}{2} dg - \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos^2 g \cdot d(-\cos g) = \\ & = \left(\frac{1}{8} \cdot 2g + \frac{1}{8} \sin 2g + \frac{1}{2} \cdot \frac{\cos^3 g}{3} \right) \Big|_{-\pi/2}^{\pi/2} = \frac{1}{4} \cdot \frac{\pi}{2} + \frac{1}{8} \sin 2 \cdot \frac{\pi}{2} + \\ & + \frac{1}{6} \cos^3 \frac{\pi}{2} - \frac{1}{4} \cdot \left(\frac{-\pi}{2} \right) - \frac{1}{8} \sin 2 \cdot \left(\frac{-\pi}{2} \right) + \frac{1}{6} \cos^3 \left(\frac{-\pi}{2} \right) = \frac{\pi}{8} + \frac{\pi}{8} = \frac{\pi}{4} \end{aligned}$$